## 1 Summary

# Waste Disposal — Vacuum Toilet Assembly

Vacuum Toilet Blockages and Flush Valve Removals on Short-Sector Operation

| | | | |
|---|---|---|---|
| **Object code** | DMC-CRN-A-38-30-11-00A-018A-A | **Issue** | 001-00 (new) |
| **Issue date** | 2025-03-17 | **Language** | en-GB |
| **Responsible partner** | AERALIS S.A.S. (U8025) | **Originator** | AERALIS S.A.S. — Customer Support Engineering, Neuhafen (U8025) |
| **Security** | 01 — Unclassified | **Quality assurance** | First verification (tabtop) |
| **Applicability** | CIRRANE-13, all aircraft | | |

### 1 Summary

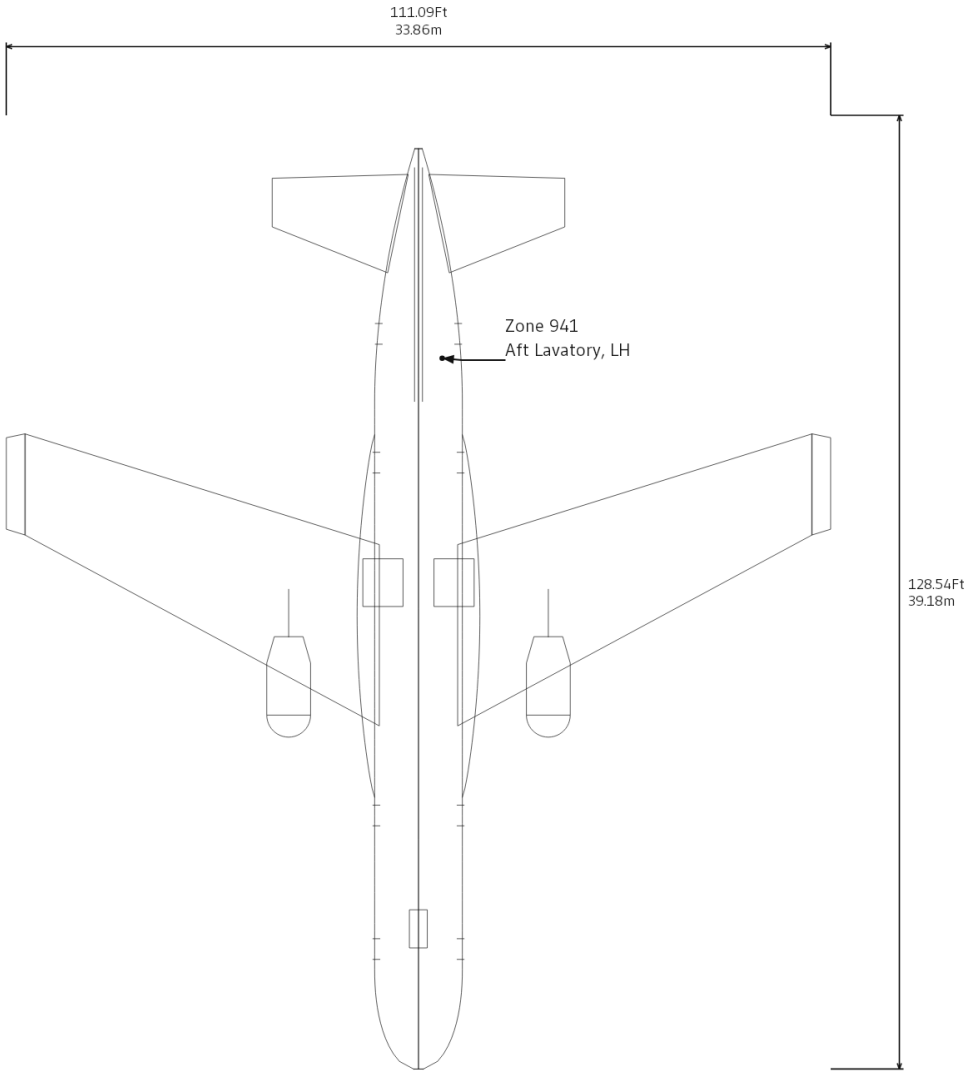
Maintenance provider reports an unusually high flush valve removal rate and repeat lavatory blockages on short-sector CIRRANE-13 aircraft, and asks for a reliability improvement rather than a further servicing instruction.

### 2 Request

We maintain a mixed CIRRANE-13 fleet for three operators and the lavatory system is now our single biggest driver of cabin defects. The flush valves come off far more often than the published rate suggests they should, and the aft lavatory blocks up in a way the forward one does not. On short sectors the crew flush more often, the tank never gets a long cruise to settle, and by the end of a day of six or seven sectors we are getting reports of slow flush or no flush. Our technicians clear the blockage, the aircraft goes out, and it comes back two days later.

We have followed the servicing recommendations to the letter, including the extra rinse cycle, and it has not changed the numbers. We would like the design looked at rather than another servicing instruction, because we cannot service our way out of this.

## 3 Background




**Waste Disposal — Vacuum Toilet Assembly — location, plan view — Figure 1**

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Across the twenty-six aircraft we look after we replaced forty-one flush valves in the last twelve months. The published mean time between unscheduled removals for the valve is about four times what we are seeing. The pattern is strongly biased to the aft lavatory and to operators flying five or more sectors a day. Two of the three operators have raised it as a cabin readiness issue with their own customers.

We have already applied the revised rinse procedure issued in 2024 across the whole fleet.

## 4 Expected outcome

## 5 Order data

A change to the flush valve or to the aft waste line routing that brings the removal rate back to the published figure, embodiable at line level or during an A-check, with no change to the water servicing procedure.

## 5 Order data

**Table 1 Order data**

| | |
|---|---|
| Order number | IO-C-250216 |
| Category | Cabin |
| Priority | Medium |
| Status | Converted to mod |
| Raised by | Mro |
| Operator | CBS |
| Contact role | Technical Services Manager |
| Aeralis focal | EYIS, Neuhafen |
| Aircraft in scope | 26 |
| Funding | Warranty |
| Budget | 180 kEUR |
| Contract | MSA-2025-CBS-0044 |
| Response requested by | 2025-07-04 |
| Embodiment requested from | 2026-04-06 |
| Modification raised | MOD-111862 |

Forty-one unscheduled flush valve removals in twelve months across twenty-six aircraft, plus the associated cabin defect reports, exceed the maintenance provider's contracted allowance and are being passed back as a cost claim.

## 6 Applicability and effectivity

CIRRANE-13, all aircraft

## 7 Referenced documents

**Table 2 Referenced documents**

| Document | Type | Relationship |
|---|---|---|
| MOD-111862 | MOD | Created |